

Epic 2: Visualizing and Analysing the Data

Story 5: Descriptive Statistical Analysis

Descriptive statistical analysis is performed to understand the overall characteristics and quality of the flood prediction dataset before applying machine learning algorithms. This analysis provides valuable information about the dataset structure, data types, statistical measures, and missing values, enabling effective preprocessing and model development.

Dataset Exploration Functions

- **head()** – Displays the first five records of the dataset, allowing quick verification of the data structure and feature values.
- **info()** – Provides information about the dataset, including the number of rows, columns, data types, and the presence of missing (null) values.
- **describe()** – Generates descriptive statistics such as count, mean, standard deviation, minimum, maximum, and quartile values (25%, 50%, and 75%) for all numerical features.
- **Data Type Analysis** – Identifies whether each feature is numerical or categorical, helping determine suitable preprocessing techniques such as encoding, scaling, or normalization.

Outcome

- Examined the overall structure of the flood prediction dataset.
- Verified data types and detected any missing values.
- Generated statistical summaries for numerical features.
- Confirmed that the dataset is suitable for preprocessing and machine learning model development.

Figure 6: Output of head(), info(), and describe() showing the structure and statistical summary of the flood prediction dataset.

```
#check the first 5 observations
dataset.head()
```

	Temp	Humidity	Cloud Cover	ANNUAL	Jan-Feb	Mar-May	Jun-Sep	Oct-Dec	avgjune	sub	flood
0	29	70	30	3248.6	73.4	386.2	2122.8	666.1	274.866667	649.9	0
1	28	75	40	3326.6	9.3	275.7	2403.4	638.2	130.300000	256.4	1
2	28	75	42	3271.2	21.7	336.3	2343.0	570.1	186.200000	308.9	0
3	29	71	44	3129.7	26.7	339.4	2398.2	365.3	366.066667	862.5	0
4	31	74	40	2741.6	23.4	378.5	1881.5	458.1	283.400000	586.9	0